IE643 Lecture 9

Announcements:

→ Project topic preference form submission deadline (tonight) 28th Aug. 11:59 pm Indian time.

→ MLP Demo - 2 to 4 pm tomorrow. Venue: to be announced by tonight.

Backward Pass (Recall)

$$\frac{\partial e}{\partial w_{ij}^l} = \frac{\partial e}{\partial z_i^l} \cdot \left(\frac{\partial z_i^l}{\partial w_{ij}^l} \right)$$

[Diagram: A chain of nodes a_j^{l-1} feeding into z_i^l, which feeds into a_i^l.]

$$\frac{\partial e}{\partial a_i^l} = \sum_m \frac{\partial e}{\partial a_m^{l+1}} \cdot \left(\frac{\partial z_m^{l+1}}{\partial a_i^l} \right)$$

$$\frac{\partial e}{\partial a_i^l} = \sum_m \frac{\partial e}{\partial a_m^{l+1}} \phi'(z_m^{l+1}) w_{mi}^l$$

Vanishing/Exploding Grad

$$\begin{bmatrix} \frac{\partial e}{\partial a_1^l} \\ \frac{\partial e}{\partial a_2^l} \\ \vdots \\ \frac{\partial e}{\partial a_{N_l}^l} \end{bmatrix} = (W^{l+1})^T \text{Diag}\left(\phi'(z^{l+1})\right)$$

Vanishing/exploding grad

$$\begin{bmatrix} \frac{\partial e}{\partial a_1^{l+1}} \\ \vdots \\ \frac{\partial e}{\partial a_{N_{l+1}}^{l+1}} \end{bmatrix}$$

MLP for multi-class case.

Input space: $X \subseteq \mathbb{R}^d$

Output space: $Y = \{1, 2, 3, \dots C\}$

$$y = \{0, 1\}, \quad \{+1, -1\}$$

$(x, y) \in X \times Y, \qquad D = \{(x^i, y^i)\}_{i=1}^N, x^i \in X, y^i \in Y$

Fully-Connected Layers

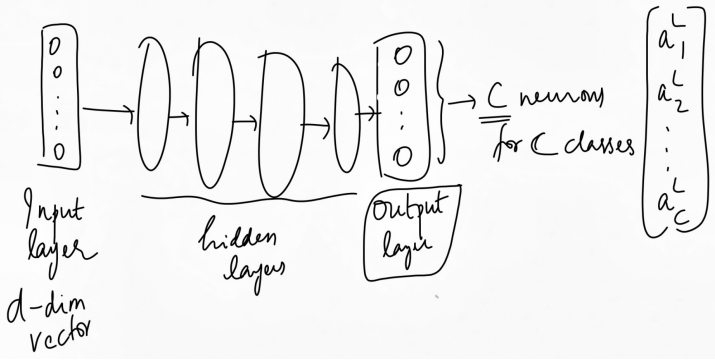


Figure 1. A chain of fully-connected layers — input layer (a d-dim vector) → several hidden layers → output layer (C neurons, one per class).

Input layer: d-dim vector

hidden layers

Output layer: C neurons for C classes

$$\begin{bmatrix} a_1^L \\ a_2^L \\ \vdots \\ a_C^L \end{bmatrix}$$

Output Layer

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_C \end{bmatrix}, \quad y \in \{1, 2, \dots C\} \rightarrow \text{Transform discrete } y \text{ into a } C - \text{dim vector.}$$

$$y = k \in \{1, 2, \dots C\} \rightarrow \text{one "hot" vector:}$$

[Diagram: A column of C-dimensional vectors, each with a single 1 in the kth position and 0s elsewhere, labeled "1st component".]

Discrete Prob Distn

$$y = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_C \end{bmatrix}, \quad \{p_j \geq 0, \ \sum_j p_j = 1\}$$

$\downarrow$

$\{p_1, p_2, \dots, p_C\}$ which form a discrete prob. distn.

where the entire mass is on the kth component.

$$\hat{y} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_C \end{bmatrix}, \quad q_j \geq 0, \ \sum q_j = 1$$

kL div b/w p and q

$$\text{kL}(p||q) = \sum_{j=1}^{C} p_j \log \frac{p_j}{q_j}$$

$$= E_p \log \frac{p}{q}$$

Compare two discrete prob distri

- KL divergence
- Bregman div
- JS div ...

$$\text{KL}(p \to q)$$

$$\text{KL}(q \to p) = \text{KL}(q \| p) = \sum_{j=1}^{C} q_j \log \frac{q_j}{p_j}$$

$$\text{KL}(p \| q) = \sum_{j=1}^{C} p_j \log p_j - \sum_{j=1}^{C} p_j \log q_j$$

$\to$ negative entropy.

$$\gamma = q \ \text{KL} \downarrow 0 \ \text{KL} \uparrow$$

Entropy

$$\text{Entropy}(p) := -\sum p_j \log p_j$$

$$p_1, p_2, \cdots, p_C$$

$$KL = N\epsilon + C\epsilon \rightarrow \text{cross-entropy}$$

$$D = \{(x^i, y^i)\}_{i=1}^N, \quad y^i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ b \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^C$$

$$\widehat{y}^i = \begin{bmatrix} \widehat{y}_1^i \\ \widehat{y}_2^i \\ \vdots \\ \widehat{y}_C^i \end{bmatrix}$$

Logistic Regression

$$c(y^i, \widehat{y}^i) = \text{KL}(y^i, \widehat{y}^i)$$

$\rightarrow$ KL = 0 when $y^i = \widehat{y}^i$, KL > 0 when $y^i \neq \widehat{y}^i$

logistic regression

$$\min_{\text{weights of NN}} \sum_{i=1}^N \text{KL}(y^i, \widehat{y}^i)$$

$$\min_{\text{weights}} \sum_{i=1}^N \text{CE}(y^i, \widehat{y}^i)$$

$$E[y|x = \vec{x}] \rightarrow \text{Bernoulli r.v. likelihood}$$

$$\min_{\text{weights}} \sum_{i=1}^N \left(-\sum_{j=1}^C y_j^i \log \widehat{y}_j^i \right)$$

$$\min_{\text{weights}} \sum_{j=1}^C \widehat{y}_j \log \frac{\widehat{y}_j}{y_j} = \sum_{j=1}^C \widehat{y}_j \log \widehat{y}_j - \sum_{j=1}^C \widehat{y}_j \log y_j$$

Soft-Max Function

[Diagram: A column of output neurons $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_C$, each connected to a separate box (layer input).]

$$\hat{y}_j > 0, \quad \sum_j \hat{y}_j = 1$$

$$a_1^L \leftarrow 0,\ a_2^L \leftarrow 0,\ \dots,\ a_C^L \leftarrow 0$$

$$\hat{y}_j = \frac{\exp(a_j^L)}{\sum_k \exp(a_k^L)}, \quad k = 1$$

Soft-max function

$$\hat{y}_j = \frac{\exp(a_j^L)}{\sum_k \exp(a_k^L)}, \quad k = 1$$

min-max scaling: 1.

Feedforward Neural Network

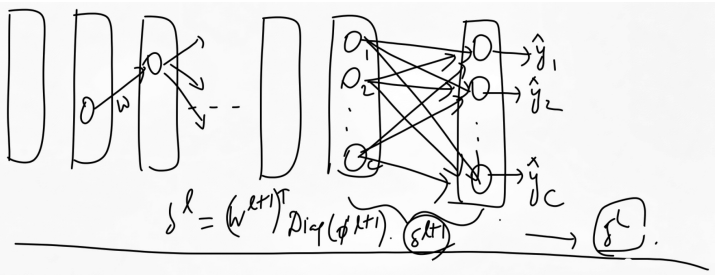


Figure 2. A chain of fully-connected layers (left) vs. a fully-connected layer feeding into a single output neuron (right).

$$\delta^l = (W^{l+1})^T \mathrm{Diag}(\phi'^{l+1}) \cdot (\delta^{l+1})$$

two pas: error is CE. bckwd pas: ??

$$\delta^L = \begin{bmatrix} \frac{\partial e}{\partial a_1^L} \\ \frac{\partial e}{\partial a_2^L} \\ \vdots \\ \frac{\partial e}{\partial a_C^L} \end{bmatrix}$$

[Diagram: A chain of neurons labeled $a_1^L, a_2^L, \dots, a_C^L$, each feeding into the next.]

$$e^i = -\sum_{j=1}^{C} y_j^i \log y_j^i$$

$$\frac{\partial e^i}{\partial a_k^L} = \sum_{k=1}^{C} \frac{\partial e^i}{\partial \hat{y}_k^i} \cdot \frac{\partial \hat{y}_k^i}{\partial a_j^L}$$

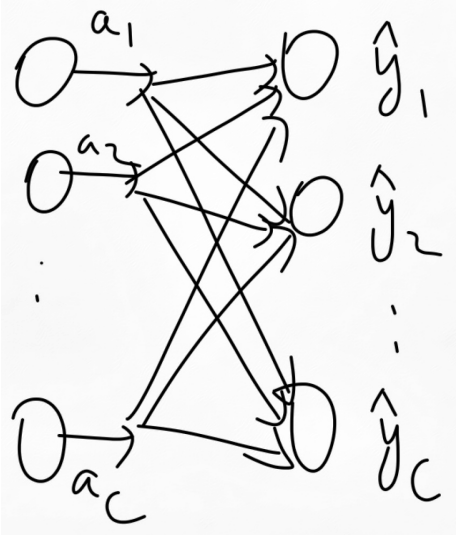


Figure 3. A feedforward neural network with input layer nodes $a_1, a_2, \ldots, a_c$, fully connecting to an output layer node (neuron) computing $\hat{y}_j$ for each of c outputs.

$$\frac{\partial e}{\partial a_j} = \sum_{k=1}^{c} \frac{\partial e}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial a_j}$$

$$= \sum_{k=1}^{c} -\frac{y_k}{\hat{y}_k} \left(\frac{\partial \hat{y}_k}{\partial a_j} \right)$$

$$e = -\sum_{j=1}^{c} y_j \log \hat{y}_j$$

$$\frac{\partial \hat{y}_k}{\partial a_j} = \begin{cases} \hat{y}_j - (\hat{y}_j)^2 & \text{when } k = j \\ -\hat{y}_k \hat{y}_j & \text{when } k \neq j \end{cases}$$

$$\hat{y}_k = \frac{\exp(a_k)}{\sum_{m=1}^{n} \exp(a_m)}$$

$$= \sum_{k \neq j} (-y_k)(-\hat{y}_j) - y_j(1 - \hat{y}_j) + \left(\frac{-y_j}{\hat{y}_j} \right) (\hat{y}_j(1 - \hat{y}_j))$$

$$= \sum_{k \neq j} y_k \hat{y}_j - y_j + y_j \hat{y}_j$$

$$e = \frac{1}{2}\sum_j (\hat{y}_j - y_j)^2$$

$$\frac{\partial e}{\partial a_j} = ??$$

$$= \sum_{k=1}^{n} y_k \hat{y}_j - y_j$$

$$= \hat{y}_j \left(\sum_{k=1}^{n} y_k \right) - y_j = \hat{y}_j - y_j = \frac{\partial e}{\partial a_j}$$

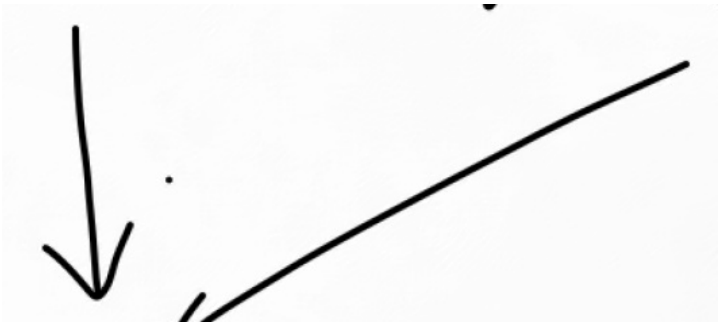


Figure 4. drawing from the page (no description in the notes)

Backward Pass

$$\begin{bmatrix} \frac{\partial e}{\partial a_1} \\ \frac{\partial e}{\partial a_2} \\ \vdots \\ \frac{\partial e}{\partial a_c} \end{bmatrix} = \begin{bmatrix} \hat{y}_1 - y_1 \\ \hat{y}_2 - y_2 \\ \vdots \\ \hat{y}_c - y_c \end{bmatrix}$$

$$= \delta$$